33 Multi-sensor Diagnostic Protocol in areas up to 3×2 m: FLOSS Workflows, Repeatability, and the Faro Convention

Matteo Orsi (m.orsi@labdig3a.org), Gloria Valbusa, Filippo di Marco, Roberto Taglioretti

Cultural heritage diagnostics require not only metric accuracy but also the absolute repeatability of measurements over time. Without a standardized protocol, multi-sensor data remain irreconcliable, unverifiable and biased by proprietary hidden processing constraints. This paper presents a preliminary methodology on areas up to 3x2 m, designed to encode a strict diagnostic protocol integrating photogrammetry (SfM), structured light scanners, and Lidar Scanner (used exclusively for point cloud generation and GNSS referencing). The protocol overcome the limitations of commercial hardware: currently, while data acquisition inevitably relies on the sensors' proprietary software, the processing of raw data is executed in a strictly FLOSS environment. This open architecture eliminates commercial "black-boxes" and "vendor lock in" ensuring that the data processing phase fully complies with the principles of democratization and accessibility written in the Faro Convention (Italian Law 133/2020). The guarantee of metric accuracy and scientific repeatability relies on a transparent and traceable pipeline integrating Meshroom (Griwodz et al. 2021; Meshroom – Node-Based Visual Programming Toolbox 2026), CloudCompare (CloudCompare - Open Source Project, n.d.), and MOSAIC (Multi-sensor Osteological & Stratigraphic Analysis for In-situ Contexts, a MAPOD4D Lab subproject under development) (Taglioretti et al. 2025). MOSAIC is essential for automating noise cleaning and implementing custom C/C++/Python nodes within Meshroom for direct dense cloud extraction. CloudCompare is employed for co-registration, metric validation, and meshing. The protocol standardizes the reproducible production of four specific deliverables: 1) SfM sensor positions with high quality photographic texture; 2) Dense point cloud for diagnostic and scientific use; 3) Mid-poly mesh for 3D printing; 4) Low-poly mesh for interactive enhancement software (e.g., MAPOD4D Multiverse). By replacing opaque algorithms with a 100% FLOSS protocol, we ensure total analytical control, high metric accuracy, and the strict replicability of the diagnostic investigation.